Lab 07: Improving HOG-Based Medical Image Classification Using a Genetic Algorithm

Objective

Modify the traditional Histogram of Oriented Gradients (HOG) feature-extraction pipeline by integrating a Genetic Algorithm (GA) for feature selection. Evaluate whether the modified approach improves classification performance on a medical image dataset.

Tasks

1. Select a suitable medical image dataset, such as BloodMNIST, PathMNIST, or PneumoniaMNIST.
2. Preprocess the images and create separate training, validation, and test sets.
3. Implement the baseline HOG + SVM classification pipeline.
4. Extract HOG features from the medical images.
5. Implement a Genetic Algorithm with population initialization, fitness evaluation, selection, crossover, and mutation.
6. Use the GA to select a subset of HOG features based on validation macro-F1 and a feature-selection penalty.
7. Train an SVM classifier using the selected features.
8. Compare the baseline and modified models using accuracy, macro-F1, balanced accuracy, and computational time.
9. Plot the GA fitness across generations and compare the number of features before and after selection.
10. Discuss the strengths, limitations, and computational cost of the proposed method.

Experimental Requirements

- Use identical data splits for all approaches.
- Compare HOG + SVM against GA-HOG + SVM.
- Experiment with at least two GA settings, such as population size or mutation probability.
- Report actual experimental results in tables and graphs.
- Perform final evaluation on the test set only after model selection is complete.

Submission

Submit the source code, dataset reference, experimental results, plots, and a short report explaining whether GA-based feature selection improved classification performance.

Research Question

Can Genetic Algorithm-based feature selection improve the classification performance of HOG features while reducing feature dimensionality on medical images?

Experimental Results and Comparison Tables

Table 1. Experimental configuration and HOG feature-extraction parameters

This table establishes the experimental settings so the baseline and proposed method can be reproduced fairly.

Parameter	Baseline: HOG + SVM	Proposed: GA-HOG + SVM
Medical dataset	Same dataset	Same dataset
Image resolution	128 × 128	128 × 128
HOG orientations	9	9
Pixels per cell	8 × 8	8 × 8
Cells per block	2 × 2	2 × 2
Block normalization	L2-Hys	L2-Hys
Feature selection	None	Genetic Algorithm
Classifier	SVM	SVM
SVM kernel	RBF	RBF
Hyperparameter tuning	Validation set	Validation set
Data split	Identical fixed split	Identical fixed split
Random seed	Fixed	Fixed

Table 2. Main performance comparison: Without GA versus With GA

Metric	HOG + SVM (Baseline)	GA-HOG + SVM (Proposed)	Improvement
Accuracy (%)	—	—	—
Precision (Macro, %)	—	—	—
Recall (Macro, %)	—	—	—
Macro-F1 (%)	—	—	—
Weighted-F1 (%)	—	—	—
Balanced accuracy (%)	—	—	—
Specificity (Macro, %)	—	—	—
Cohen's kappa	—	—	—
Matthews correlation coefficient	—	—	—

For accuracy, precision, recall, and F1, higher values generally indicate better performance. Specificity should be calculated using a clearly stated multiclass averaging method where applicable.